

The Interpolated Stochastic Heat and Wave Equation: Solvability and Exact Moment Asymptotics

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The Stochastic Heat and Wave Equations

$$\begin{cases} (\partial_t^b - \Delta) u(t, x) = \sqrt{\theta} u(t, x) \dot{W}(x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b = 1 \quad (\text{SHE}) \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b = 2 \quad (\text{SWE}) \end{cases}$$

- ▶ $W = \{W(\phi) : \phi \in \mathcal{D}(\mathbb{R}^d)\}$ is a centered and time-independent Gaussian noise
- ▶ The choice of initial condition is such that the solution to the homogeneous equation is constant one.
- ▶ The solution is understood in the *Skorohod* sense.

For the SHE, i.e., $b = 1$ ($a = 2, \nu = 2$)
[X. Chen '17]

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{2-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{2}{2-\alpha}} (2-\alpha) \left(\frac{2\mathcal{M}}{4-\alpha} \right)^{\frac{4-\alpha}{2-\alpha}}$$

For the SWE, i.e., $b = 2$ ($a = 2, \nu = 2$)
[Balan, L. Chen, and X. Chen '21]

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{3-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{1}{3-\alpha}} \left(\frac{1}{2} \right)^{\frac{\alpha}{2(3-\alpha)}} \frac{3-\alpha}{2} \left(\frac{2\mathcal{M}^{1/2}}{4-\alpha} \right)^{\frac{4-\alpha}{3-\alpha}}$$

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$$b \in (1, 2) ??$$

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The Interpolated Stochastic Heat and Wave Equation.

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases}$$

- ▶ ∂_t^b is the *Caputo* fractional derivative
- ▶ $(-\Delta)^{a/2}$ is the fractional Laplacian of order $a \in (0, 2]$
- ▶ I_t^r is the Riemann-Liouville fractional integral of order $r \geq 0$

The Noise, $\dot{W}$.

The time independent noise informally satisfies

$$\mathbb{E}(\dot{W}(x)\dot{W}(y)) = \gamma(x - y).$$

The spatial correlation and spectral density, γ and φ , for the noise $\dot{W}$ can be assumed to be any of the following:

$$\gamma(x) = |x|^{-\alpha}, \quad \varphi(\xi) = C_1 |\xi|^{d-\alpha} \quad \alpha \in (0, d)$$

$$\gamma(x) = \prod_{i=1}^d |x_i|^{-\alpha_i}, \quad \varphi(\xi) = C_2 \prod_{i=1}^d |\xi_i|^{1-\alpha_i} \quad \alpha_i \in (0, 1), \quad \alpha = \sum_i \alpha_i$$

The fundamental solution.

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r[f(t, x)] & x \in \mathbb{R}^d, t > 0, \\ u(0, \cdot) = 1 & b \in (0, 1], \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2), \end{cases}$$

When $f(t, x)$ is a deterministic function, the fundamental solution

$$G(t, x) := G_{a,b,r,\nu,d}(t, x)$$

is defined explicitly through the Fox-H function. We have for both $b \in (0, 1]$ and $b \in (1, 2)$ that

$$u(t, x) = 1 + (G \star f)(t, x)$$

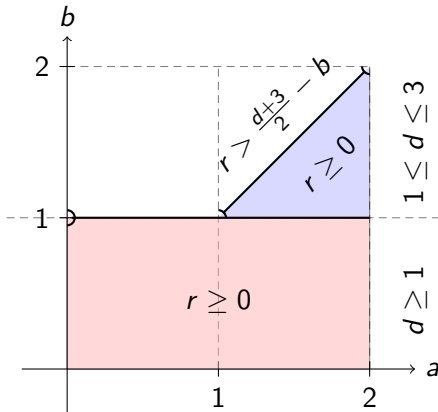
An important characteristic of G is that

$$\mathcal{FL}G(s, \xi) = \frac{s^{-r}}{s^b + \frac{\nu}{2}|\xi|^a}$$

Nonnegativity assumption on G

Under any of the following cases, G is nonnegative [Chen, Hu, Nualart '19]:

- ▶ $d \geq 1$, $b \in (0, 1]$, $a \in (0, 2]$, $r \geq 0$;
- ▶ $1 \leq d \leq 3$, $1 < b < a \leq 2$, $r > 0$;
- ▶ $1 \leq d \leq 3$, $1 < b = a < 2$, $r > \frac{d+3}{2} - b$.



Mild solution

Definition

For $T > 0$, a random field $u = \{u(t, x) : t \in (0, T), x \in \mathbb{R}^d\}$ is called a *mild solution* if $G(t - s, x - \cdot)u(s, \cdot)1_{\{s < t\}}$ is Skorohod integrable and the following holds almost surely:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t - s, x - y) u(s, y) W(\delta y) \right) ds.$$

- ▶ *Global Solution:* $\|u(t, x)\|_p < \infty$ for any $t > 0$ and $x \in \mathbb{R}^d$.
- ▶ *Local Solution:* If there exists $0 < T_a \leq T_b < \infty$ such that $\|u(t, x)\|_2 < \infty$ when $0 < t < T_a$ and $\|u(t, x)\|_2$ D.N.E. for $t > T_b$.

Wiener Chaos Expansion:

Through a standard procedure, we may define

$$f_n(x_1, \dots, x_n; x, t) = \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} \prod_{k=1}^n G(t_{k+1} - t_k, x_{k+1} - x_k) dt_1 \cdots dt_n$$

where $t = t_{n+1}$ and $x = x_{n+1}$, and say that

$$u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)), \quad (t, x) \in (0, T) \times \mathbb{R}^d$$

and

$$\mathbb{E}(u(t, x)^2) = \sum_{k=0}^{\infty} \theta^k \|f_k(\cdot, x, t)\|_{\mathcal{H}^{\otimes n}}^2, \quad (t, x) \in (0, T) \times \mathbb{R}^d.$$

Theorem 1.6 (Chen and Eisenberg '21+)

Assuming $G(t, x)$ is nonnegative and the noise has spatial correlation has the form $\gamma(x) = \prod_{i=1}^k |x_{(i)}|^{-\alpha_i}$:

► A *global solution* exists provided

$$0 < \alpha < \min \left(\frac{a}{b} [2(b+r) - 1], 2a, d \right)$$

► Otherwise, if

$$r \in [0, 1/2] \quad \text{and} \quad 0 < \alpha = \frac{a}{b} [2(b+r) - 1] \leq d,$$

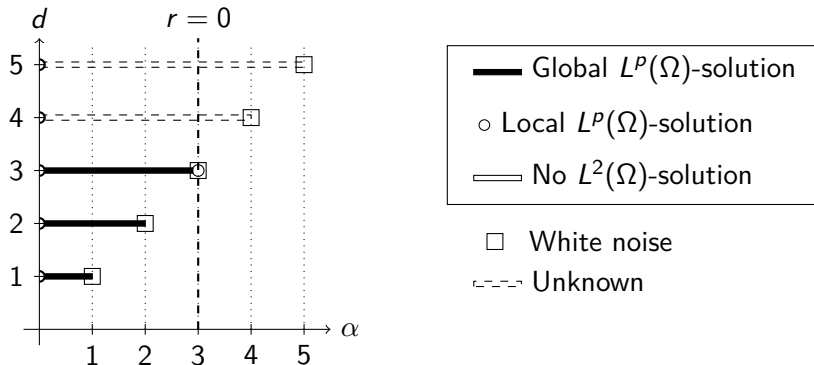
then a *local solution* exists for $t \in (0, T_p)$ where

$$T_p := \frac{\nu^{\alpha/a}}{2\theta(p-1)\mathcal{M}_a^{(2a-\alpha)/a}}$$

and the solution D.N.E. for $t > T_2$.

Example: Solvability for the SWE ($a = b = 2$)

A local solution only exists when $\alpha = 3 + 2r \leq d \leq 3$.

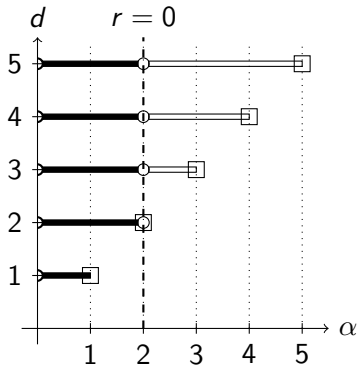


By replacing $\nu = 2$ in the following, we recover (1.12) Balan, Chen and Chen:

$$T_p = \frac{\nu^{3/2}}{2\theta(p-1)\sqrt{\mathcal{M}_{2,3}(\delta_0)}}, \quad p \geq 2$$

Example: Solvability for the SHE ($a = 2$ and $b = 1$)

By setting $a = 2$, $b = 1$ and $r = 0$, we obtain the following condition for existence of a local solution: $\alpha = 2 \leq d$.



- ▶ When $\alpha = d = 2$, the critical time becomes
$$T_p = \frac{\nu}{2\theta(p-1)\mathcal{M}_{2,2}(\delta_0)}.$$
- ▶ Theorem 4.1 (Hu) proves that an $L^2(\Omega)$ solution exists for $t < 2$ but not for $t > 2\pi$.
- ▶ T_2 being precise implies
$$2 \leq T_2 = \frac{1}{2\mathcal{M}_{2,2}(\delta_0)} \leq 2\pi$$

How to find the critical α .

$$\begin{aligned}\|u(t, x)\|_2^2 &= \sum_{n \geq 0} \theta^n n! \overbrace{t^{[2(b+r)-b\alpha/a]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2}^{\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2} \\ &= \sum_{n \geq 0} \frac{\theta^n t^{(2(b+r)-b\alpha/a)n}}{(n!)^{2(b+r)-(b\alpha/a)-1}} R_n\end{aligned}$$

and $R_n = (n!)^{2(b+r)-b\alpha/a} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2$. When $\alpha = \frac{a}{b}[2(b+r)-1]$, then the above reduces down to

$$\|u(t, x)\|_2^2 = \sum_{n \geq 0} (\theta t)^n n! \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2$$

and we lose the $n!$ term in the demonstrator.

ρ

We define

$$\rho_{\nu,a}(\gamma) = \sup_{\|f\|_{L^2(\mathbb{R}^d)}=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx)$$

Theorem (Theorem 2.3 X. Chen '07)

When $\mu(dx) = \varphi(x)dx$ where φ is the generalized Riesz kernel introduced above then,

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} \left(\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} |\sum_{j=k}^n \xi_{\sigma(j)}|^a} \right)^2 \mu(d\vec{\xi}) \right] \\ = \log(\rho_{\nu,a}(\gamma)). \end{aligned}$$

where γ is the spatial correlation function.

Connection of the SPDE with ρ

Recall from above that

$$\mathcal{FLG}(1, \xi) = \frac{1}{1 + \frac{\nu}{2} |\xi|^a}.$$

By using this relation, it can be shown that

$$\mathcal{FL}(\tilde{f}_n)(1, \xi) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_{\sigma(j)} \right|^a}.$$

Applying this to the limit representation of ρ above yields that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\int_{(\mathbb{R}^d)^n} \left| \mathcal{FL}(\tilde{f}_n)(1, \xi) \right|^2 \mu(d\vec{\xi}) \right] = \log(\rho_{\nu, a}(\gamma)).$$

Lemma for the Blowup Time (SWE)

For simplicity, consider let's consider the SWE ($a = b = 2$ and $r = 0$.) The following scaling property can be easily derived by a change of variables.

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^{(4-\alpha)n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2.$$

Using this scaling property we see that

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \Gamma((4-\alpha)n + 1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2.$$

The following lemma is also very important. It is how ρ is introduced to the paper.

Theorem (Lemma 4.1 (R. Balan, L. Chen, X. Chen.))

- ▶ $\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma((4-\alpha)n + 1) \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log [2^{4-\alpha} \rho]$
- ▶ $\lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{4-\alpha} \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[\left(\frac{2}{4-\alpha} \right)^{4-\alpha} \rho \right]$

How to Find the Blowup Time (SWE)

For simplicity, we consider the blowup time for the second moment. Because of the Wiener-chaos representation of the solution and a scaling property of $\left\| \tilde{f}(\cdot, 0, t) \right\|_{H^{\otimes n}}^2$, we can say that for $d = \alpha = 3$,

$$\begin{aligned} \|u(t, x)\|_2^2 &= \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 \\ &= \sum_{n \geq 0} (\theta t)^n n! \left\| \tilde{f}(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2 \\ &=: \sum_{n \geq 0} (\theta t)^n R_n, \quad R_n = n! \left\| \tilde{f}(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2. \end{aligned}$$

The above lemma says that $\frac{1}{n} \log R_n \rightarrow \log(2\rho_c)$. Now the Cauchy-Hadamard theorem can be directly applied to see that the radius of convergence is precisely $(2\theta\rho_c)^{-1} = T_2$.

$\mathcal{M}$

We define

$$\begin{aligned}\mathcal{M}_{a,d}(\gamma, \theta) &:= \sup_{g \in \mathcal{F}_a} \left\{ \left(\iint_{\mathbb{R}^{2d}} g^2(x) g^2(y) \gamma(x+y) dx dy \right)^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\} \\ &= \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * g^2, \gamma \rangle_{L^2(\mathbb{R}^d)}^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\},\end{aligned}$$

where

$$\mathcal{E}_a(g, g) := (2\pi)^{-d} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi \quad \text{and} \quad (0.1)$$

$$\mathcal{F}_a := \left\{ f \in L^2(\mathbb{R}^d) : \|f\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(f, f) < \infty \right\}. \quad (0.2)$$

The Connection between ρ and $\mathcal{M}$

Recall that

$$\rho_{\nu,a}(\gamma) = \sup_{\|f\|_{L^2(\mathbb{R}^d)}=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(x+y)f(y)}{\sqrt{1+\frac{\nu}{2}|x+y|^a} \sqrt{1+\frac{\nu}{2}|y|^a}} dy \right]^2 \mu(dx)$$

and

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left[\int_{(\mathbb{R}^d)^n} \left| \mathcal{FL}(\tilde{f}_n)(1, \xi) \right|^2 \mu(d\xi) \right] = \log(\rho_{\nu,a}(\gamma)).$$

Theorem (X. Chen (2007))

When $\mu(dx) = \varphi(x)dx$ where φ is the generalized Riesz kernel introduced above then,

$$\rho_{\nu,a}(\gamma) = \nu^{-\alpha/a} \mathcal{M}_a^{2-(\alpha/a)}(\gamma) < \infty$$

The moment asymptotic

Theorem

Suppose a global solution to the SPDE in question exists and $\dot{W}$ is given through the generalized Riesz kernel defined above. Then,

$$\begin{aligned} & \lim_{t_p \rightarrow \infty} t_p^{-\beta} \log \|u(t, x)\|_p \\ &= \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^\beta \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right), \end{aligned}$$

where

$$\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1} \quad \text{and} \quad t_p := (p-1)^{1-1/\beta} t. \quad (0.3)$$

Exponential Growth of the Moments in Time

Theorem

For $p \geq 2$ fixed, we have that

$$\begin{aligned} & \lim_{t \rightarrow \infty} t^{-\beta} \log \mathbb{E}(|u(t, x)|^p) \\ &= p(p-1)^{\frac{1}{2(b+r) - \frac{b\alpha}{a} - 1}} \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^{\beta} \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \end{aligned}$$

From this we can deduce that $t \mapsto \mathbb{E}(|u(t, x)|^p)$ will grow like $\exp(K_1 t^\beta)$, in other words, the moments will grow exponentially in time.

Exponential Growth of the moments for Fixed Time

Theorem

For $t > 0$ fixed , we have that

$$\begin{aligned} & \lim_{p \rightarrow \infty} p^{-\beta} \log \mathbb{E} (|u(t, x)|^p) \\ &= t^\beta \left(\frac{1}{2} \right) \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right). \end{aligned}$$

From this we can deduce that $p \mapsto \mathbb{E}(|u(t, x)|^p)$ will grow like $\exp(K_2 p^\beta)$, in other words, the moments will grow exponentially in p as well.

End

Thank you for listening!

Anticipated Question: Finiteness of ρ

Recall $\varphi(x) = \prod_{i=1}^n |x_{(i)}|^{d_i - \alpha_i}$. By examining Lemma 1.6 (X. Chen 2007), to show ρ is finite for the generalized Riesz kernel, then we need to prove the following:

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y) G(x) \varphi(x - y) dy dx \leq C \|F\|_{2d/(d+\alpha)} \|G\|_{2d/(d+\alpha)}$$

This can be done by using weak Young's inequality

$$\left| \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} a(x) b(x - y) c(y) dx dy \right| \leq K_{p,q,r,d} \|a\|_p \|b\|_{q,w} \|c\|_r$$

where $q^{-1} + q'^{-1} = 1$ and $p^{-1} + q^{-1} + r^{-1} = 2$ and

$$\|b\|_{q,w} = \sup_A |A|^{-1/q'} \int_A |b(x)| dx, \quad |A| < \infty.$$

We apply this with

$$a = F, \quad b = \varphi, \quad c = G$$

and we show that $\|\varphi\|_{q,w} < \infty$ with $q = d/(d - \alpha)$.

Anticipated Question: Why does ρ appear in the Lemma.

For simplicity we will consider the second moments of stochastic wave equation ($a = b = 2$ and $r = 0$). We will also only consider the limit

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma((4 - \alpha)n + 1) \left\| \tilde{f}_n(\cdot, 0; 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log [2^{4-\alpha} \rho]$$

Recall that

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt = \Gamma((4 - \alpha)n + 1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2.$$

Define $T_n := \int_{(\mathbb{R}^d)^n} \left| \mathcal{FL} \tilde{f}_n(1, \xi) \right|^2 \mu(d\xi)$. Through direct calculation and Sterlings formula:

$$\frac{c_\alpha}{C_n} 2^{(4-\alpha)n} T_n \leq \int_0^\infty e^{-t} \left\| \tilde{f}(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 dt \leq 2^{(4-\alpha)n} T_n$$

Where $\log(C_n)/n \rightarrow 0$ and $c_\alpha > 0$.

Anticipated Question: Finding the moment asymptotic

$$\mathbb{E}(|u(t, x)|^2) = \sum_{n \geq 0} z_n R_n t^{(2(b+r)-b\alpha/a)n}$$

$$R_n = (n!)^{2(b+r)-b\alpha/a} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \quad \text{and} \quad z_n = \frac{\theta^n}{(n!)^{2(b+r)-(b\alpha/a)-1}}.$$

$$\frac{1}{n} \log(R_n) \rightarrow \log \left(\left(\frac{2}{2(b+r) - \frac{b\alpha}{a}} \right)^{2(b+r) - \frac{b\alpha}{a}} \rho \right) \quad \text{as } n \rightarrow \infty.$$

If we find a β and A so that

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} z_n R_n \left(t^{(2(b+r)-b\alpha/a)} \right)^n = A.$$

then this implies

$$\lim_{t \rightarrow \infty} \frac{1}{t^\beta} \log \sum_{n \geq 0} z_n R_n \left(t^{(2(b+r)-b\alpha/a)} \right)^n = A.$$